flyPad ground: taxi route, ATC frequencies and GSX

A taxi route page on the Navigraph airport map (with green flags on the A380X OANS), the airport ATC frequencies on it, the Services page linked to GSX, and the GSX external power sync fixed when it is enabled; the taxi route now reads real ATC clearances, and the external power follows the sim's ground power

Aircraft	A32NX and A380X (the OANS flags are A380X only)
Where to find it	flyPad > **Ground** > **Taxi** and **Services**; flyPad > Settings > 3rd Party Options; A380X ND (OANS); overhead ELEC panel EXT PWR
Branch	`core/efb-ground`, based on `core/efb-performance` (stacked: needs `core/surveillance`, `core/a380x-mfd-fms` and `core/efb-performance` merged first; the taxi page uses the runway exits of the landing calculator). `core/efb-redesign` is stacked on it.
Status	Taxi route tested in the sim by the user at RJTT and CYUL, and on both aircraft with the A380X OANS flags: OK. ATC frequencies: OK. GSX power sync: OK. GSX services link: first version tested (four bugs found); the fixes of those bugs on this branch are not sim-tested here (the reworked version on core/efb-redesign was sim-tested). Added (not sim-tested, unit tests only): taxi route along real ATC clearances, GSX power sync setting re-read, external power following the sim ground power.
Pictures	Taxi pages in the local browser harness, running the build installed in the sim, on a small test airport (XTST) made for the pictures; in the sim, the page uses the Navigraph map of the real airport. No pictures for the frequencies and GSX (they need the sim data and GSX).

1. Taxi route page
2. A380X: the accepted taxi route on the OANS
3. Airport ATC frequencies on the taxi page
4. Services page linked to GSX
5. GSX external power sync when it is enabled
6. Taxi route along real ATC clearances
7. GSX power sync: the setting is re-read, so enabling it takes effect
8. External power follows the sim's ground power (GSX GPU, sim GPU, jetway)

1. Taxi route page

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Taxi | **Status:** Tested in the sim by the user at RJTT and Montreal CYUL (the fixes in "Locate the aircraft..."), and on both aircraft: OK. Unit tests of the route finder.

What it is for

Before taxiing, the crew gets a taxi clearance from ATC such as "*taxi to holding point runway 09 via M, K, A*". On an unknown airport, following it on the chart is where mistakes happen: a wrong turn, a runway crossed without clearance. This page draws the route of the clearance on the airport map, lists the runways it crosses and stops it at the runway holding point.

The real A380 has no taxi routing: its OANS (the airport map on the ND) only lets the crew place green flags and white crosses (FCOM DSC-34-10-70-20), and MSFS has no datalink taxi clearance (D-TAXI). So the route is built on the flyPad, and the cockpit displays stay as the FCOM describes them. On the A380, the accepted route is marked on the OANS with those green flags (see "A380X: the accepted taxi route on the OANS" below).

- **Departure** (default): from a gate, a ramp or cargo stand, or the aircraft position, to a runway entry.
- **Arrival**: from the runway exit you plan to take (or the aircraft position) to your gate.
- The page **suggests** the shortest route; you type the taxiways of the ATC clearance and **accept** it.

Before you start

- **Navigraph account** linked in the flyPad settings (third party options). The airport map comes from the Navigraph airport database, the same data as the A380 OANS. Without it the page says "*No airport database*".
- **Airport:** type the ICAO code, or press **ORIG** / **DEST** to take the airports of the SimBrief flight plan loaded in the flyPad. The line below shows how many gates, ramp and cargo stands the map has.
- An airport that is not in the Navigraph airport database shows "*Not in the Navigraph airport database*".

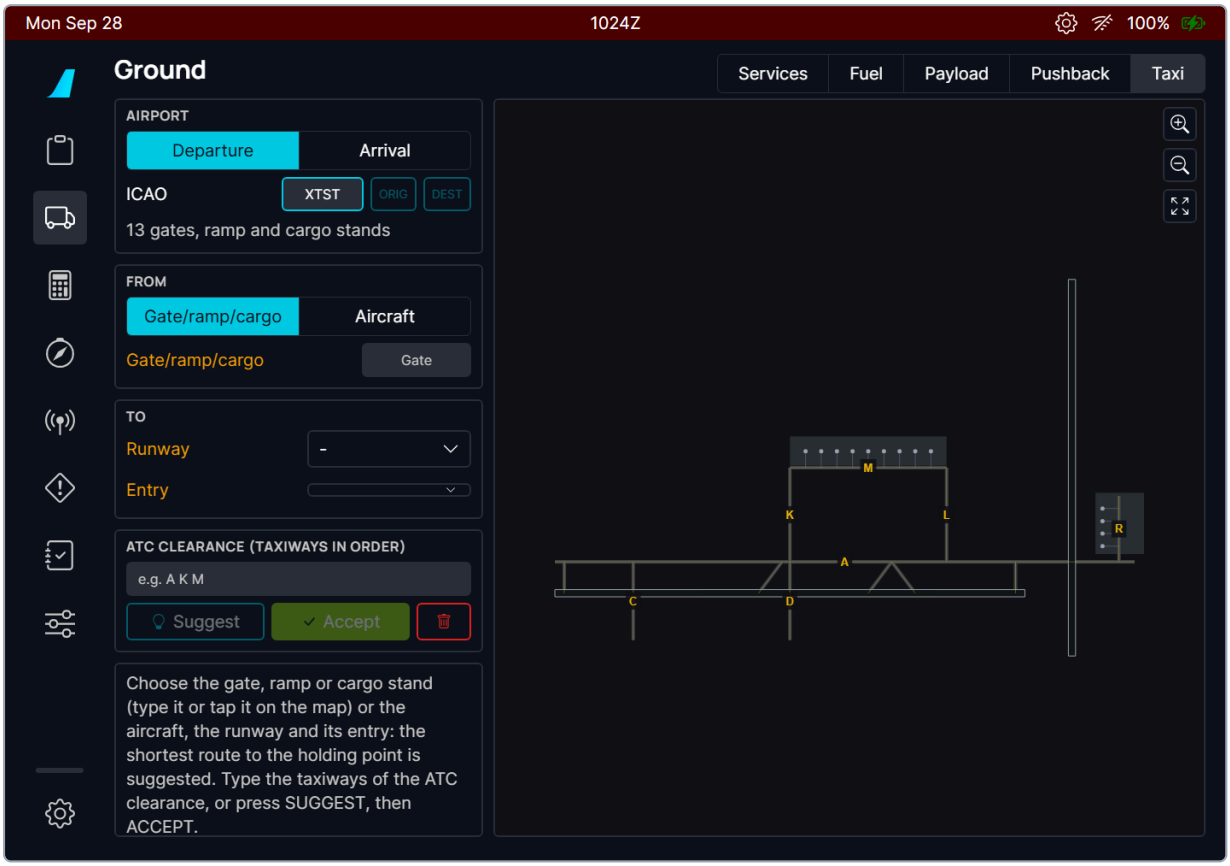


Figure 1. The Taxi page after entering the airport: the airport map on the right, the route inputs on the left. Amber labels show what is still missing.

Part of the page	What it does
AIRPORT	Departure or Arrival, the ICAO code, ORIG / DEST buttons, the number of stands on the map.
FROM	Where the taxi starts: a gate / ramp / cargo stand (departure), a runway exit (arrival), or the aircraft.
TO	Where it ends: the runway and its entry (departure), or the gate (arrival).
ATC CLEARANCE	The taxiways of the clearance, in order. SUGGEST fills it with the shortest route, ACCEPT confirms it, the red bin clears the page.
Route box	The route as a clearance reads, its length, the taxiways and the runway crossings in red.
Map	The airport with the route, the runway entries (cyan), the holding point (amber bar), the aircraft (yellow). Drag to move, buttons to zoom and to fit the route.

Departure: from the gate to the runway

1. Keep **Departure** selected and enter the airport.
2. In FROM, keep **Gate/ramp/cargo** and type your gate. While you type, the gates that match appear as buttons under the field: tap one. You can also **tap the gate on the map** (zoom in to see the names).

3. In TO, choose the **runway**. The **entry** is set to the full length entry; choose another one for an intersection departure.
4. The shortest route appears on the map as a **dashed magenta line** ending at the **HOLD** bar of the runway holding point.

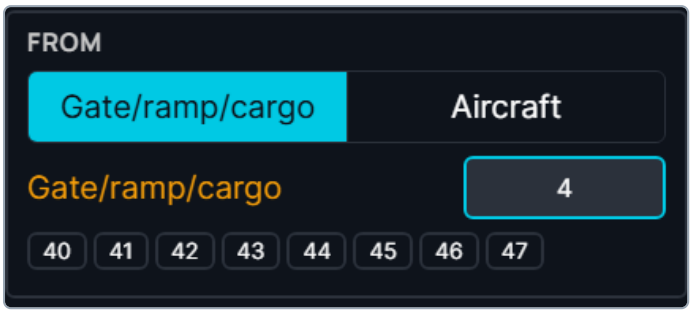


Figure 2. Typing "4" lists the gates 40 to 47: tap one.

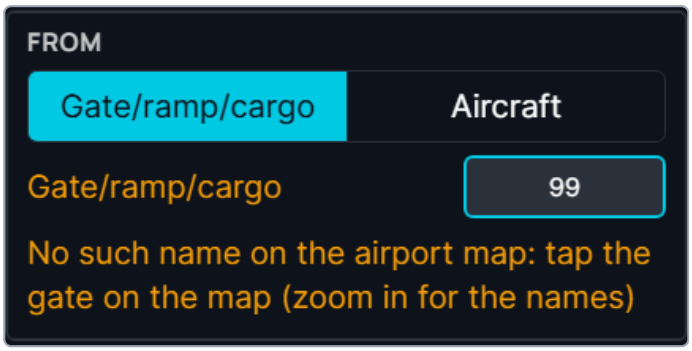


Figure 3. A name that is not on the map: the amber line says so.

Gate names are found even when they are written differently on the map: **"16" finds "016"**, spaces and dashes do not matter. If nothing matches, the airport map uses another name (for example a letter prefix): tap the gate on the map instead.

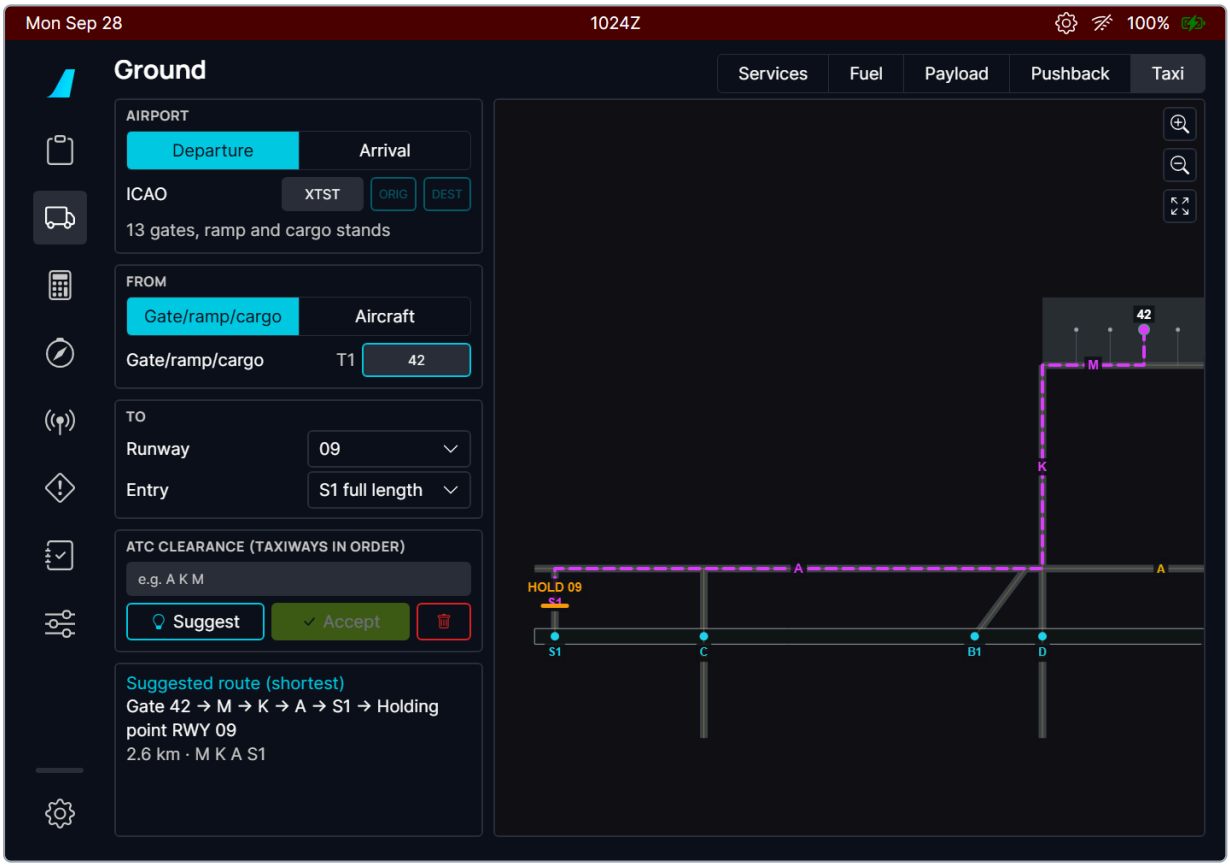


Figure 4. Gate 42 to runway 09, full length: the suggested route (dashed) along M, K and A to entry S1, stopped at the holding point (HOLD 09). The route box reads it as a clearance: Gate 42 → M → K → A → S1 → Holding point RWY 09, 2.6 km.

Runway entries

The **Entry** list shows every taxiway that enters the runway, in order from the threshold. The first one is the **full length** entry; the others show the **runway length remaining** from them, to check an intersection departure against your takeoff performance. On the map, the entries of the selected runway are the cyan dots with their names.

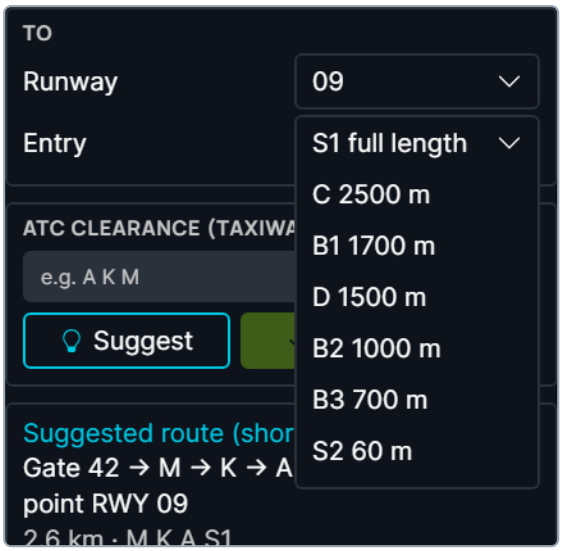


Figure 5. The entries of runway 09: S1 full length, then C (2500 m left), B1 (1700 m), D (1500 m) and so on.

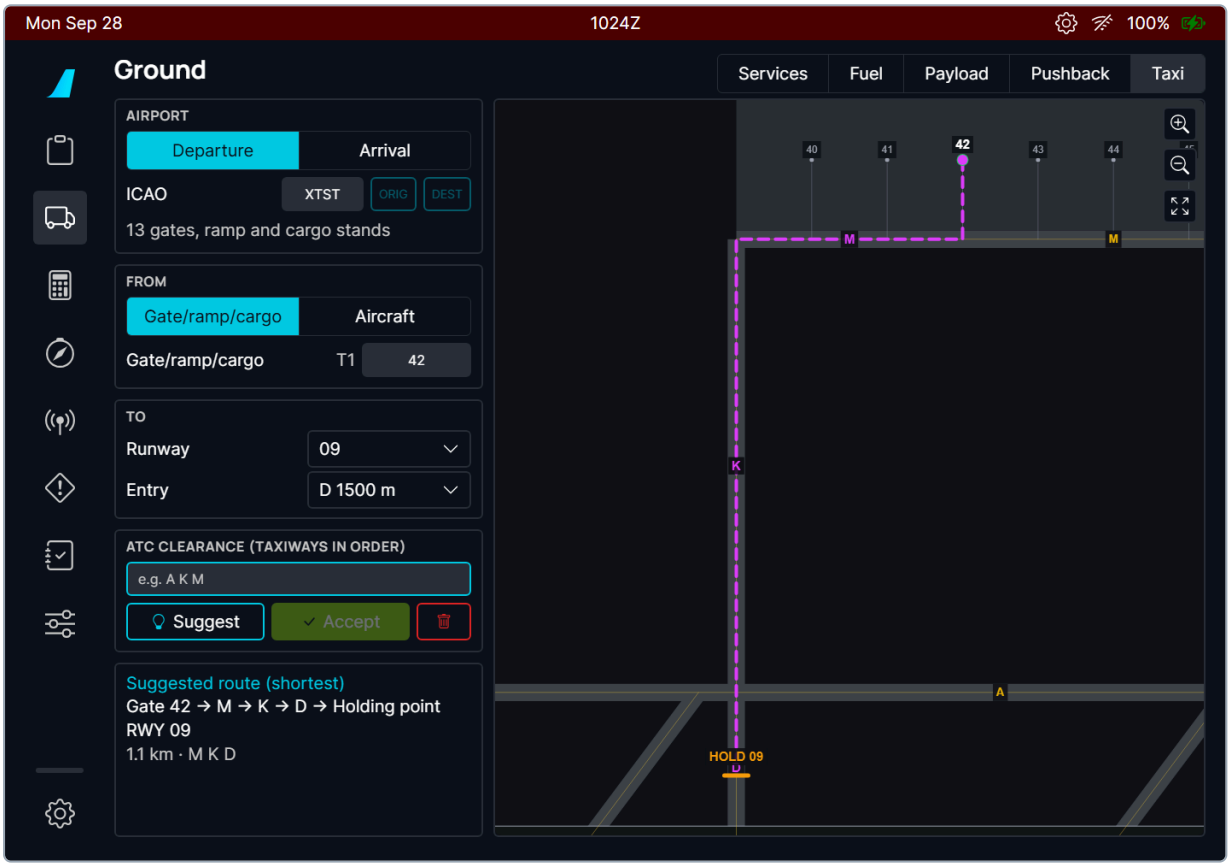


Figure 6. Intersection departure from D: the route stops at the holding point of D, with 1500 m of runway ahead.

The ATC clearance

Type the taxiways of your clearance in the order ATC gives them, separated by spaces (commas and dashes work too). The route then follows them strictly, in that order:

- Short unnamed connecting lines between two cleared taxiways are allowed.
- The runway entry can be left out: "*via M, K, A*" to a holding point still ends on the entry you chose.
- While the typed route is not accepted, the box reads "*Route along the clearance (not accepted)*" and the line stays dashed.

SUGGEST copies the taxiways of the suggested route into the clearance field: a quick start when ATC clears you along the obvious way. **ACCEPT** confirms the route: the line turns **solid**. Any change (gate, runway, entry, clearance) takes the acceptance away.

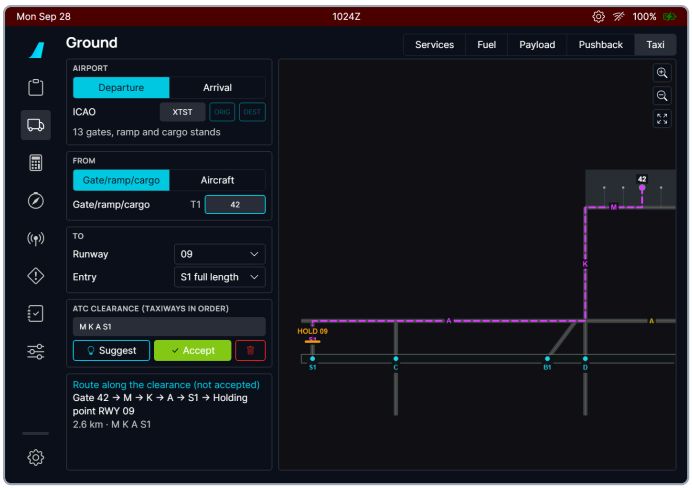


Figure 7. After SUGGEST: the clearance field holds M K A S1, the route is a preview.

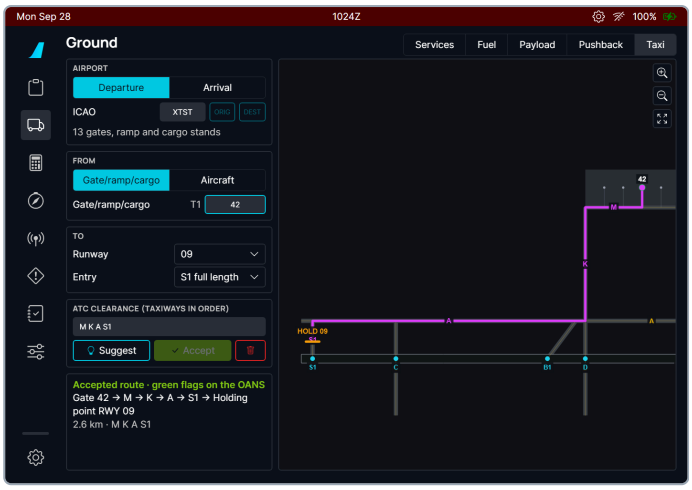


Figure 8. After ACCEPT: solid line, "Accepted route". On the A380 the route is also flagged on the OANS.

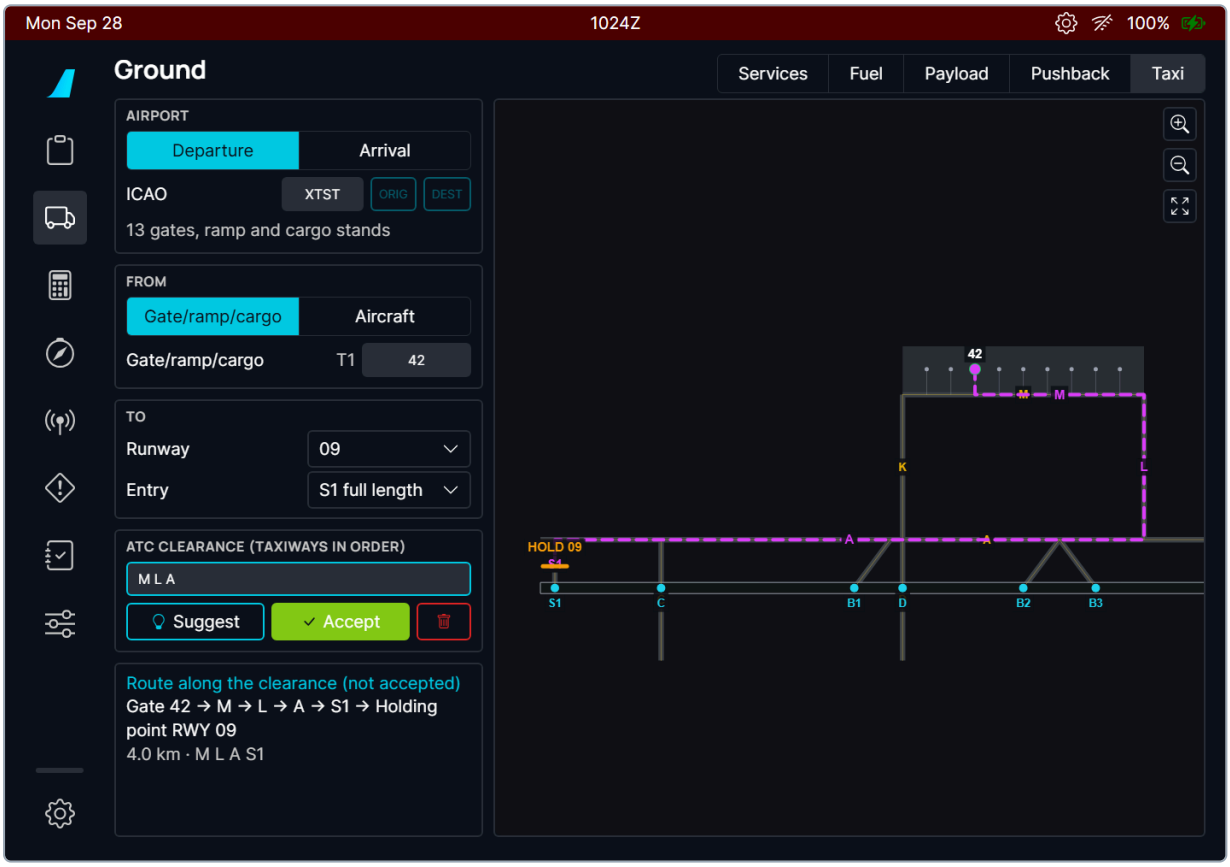


Figure 9. A different clearance, "M L A": the route goes the long way round via L (4.0 km instead of 2.6 km), as ATC asked.

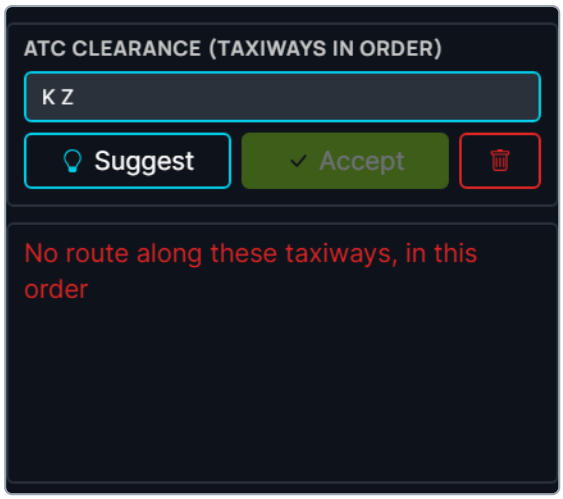


Figure 10. A clearance with a taxiway that does not exist here, or taxiways that do not connect in that order: "No route along these taxiways, in this order". Check the spelling and the order.

Runway crossings

Every runway the route crosses is circled in **red** on the map with its name, and listed in red under the route ("*Crossing runway 01/19*"). A runway crossing always needs its own ATC clearance: stop at the holding point before it and ask if you do not have it.

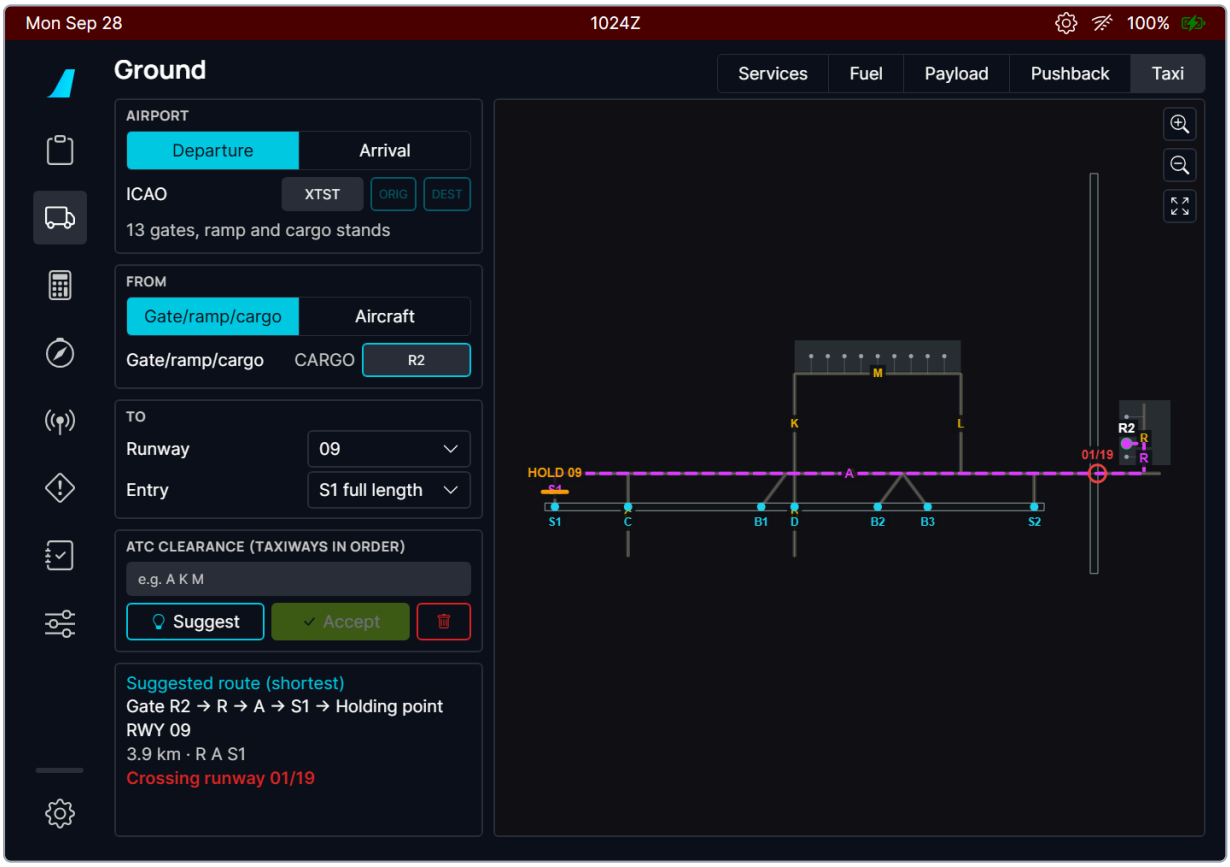


Figure 11. From cargo stand R2 to runway 09: taxiway A crosses runway 01/19 (red circle, red line in the route box).

Arrival: from the runway exit to the gate

1. Select **Arrival** and enter the airport (DEST fills it from the flight plan).
2. In FROM, keep **Runway exit**, choose the landing runway and the exit you plan to take. The exits are those of the landing calculator, with their distance from the threshold.
3. In TO, type or tap the gate.
4. Check the route, type the clearance if ATC gives another one, and ACCEPT.

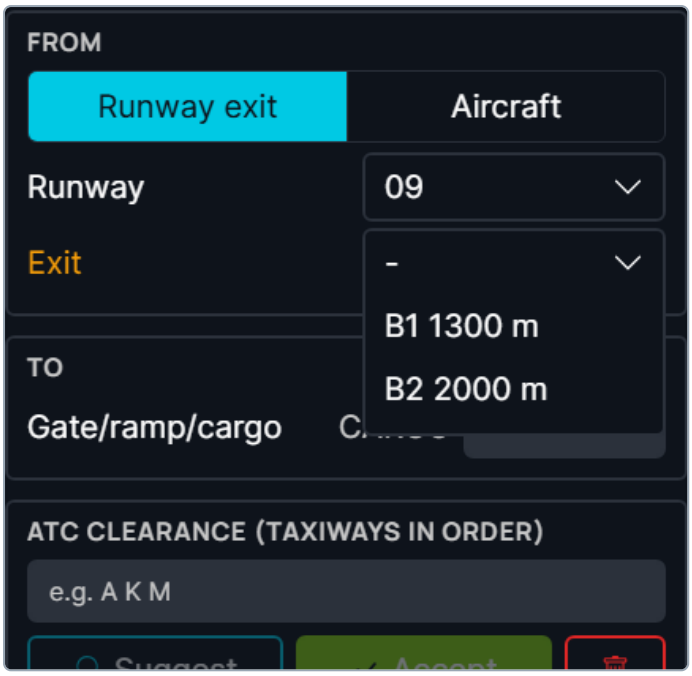


Figure 12. The exits of runway 09 with their distance from the threshold.

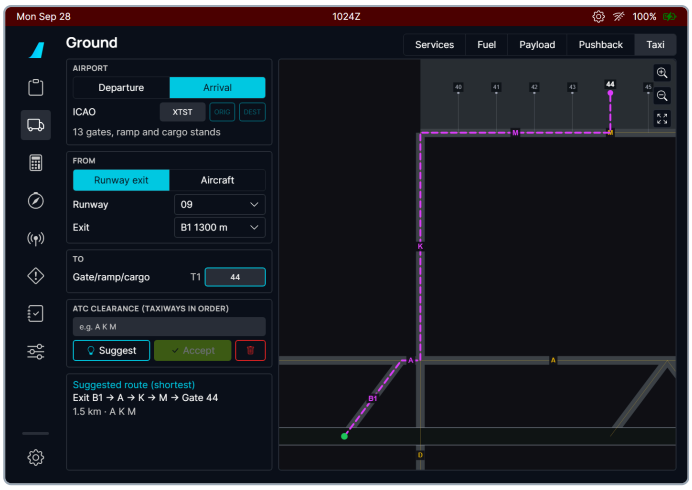


Figure 13. Exit B1 to gate 44: Exit B1 → A → K → M → Gate 44. The green dot is the start.

> **TIP**
> Prepare the arrival route during the descent briefing with the exit from the landing calculator: after landing, the page is ready. If you vacate by another exit, choose it (or use the aircraft position) and the route updates.

From the aircraft position

Choose **Aircraft** in FROM to start the route where the aircraft is, for example after a pushback, or when ATC changes the route while you taxi. The page takes the position when you select it; press **Update** to take it again. The route starts on the taxi line nearest to the aircraft and **prefers its heading**: it does not start by turning back if there is a way ahead.

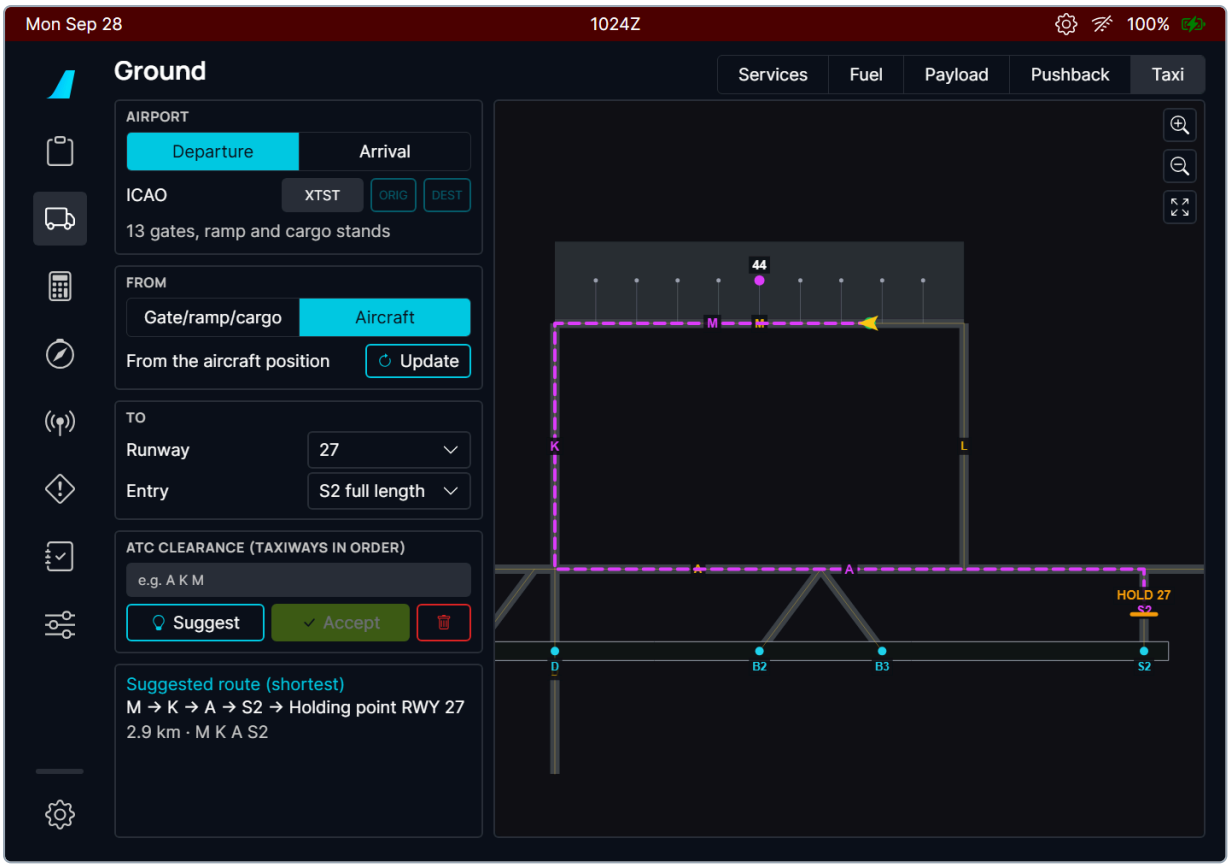


Figure 14. The aircraft (yellow) on M facing west, for runway 27: the route goes ahead via K and A (2.9 km) rather than turning back on M.

"*The aircraft is not at this airport*" means the aircraft is more than 8 km from the airport of the page.

Locate the aircraft, taxiway names of several words, proposed entry

- **Locate button** (pin, under the zoom buttons): centres the map on the aircraft and zooms in until the stand names show. It is greyed when the aircraft is not at this airport.
- Taxiways named with **several words** (e.g. SOUTH RAMP, NORTH RAMP at Montreal CYUL) are read as one taxiway in the clearance: type them as the map shows them, or press SUGGEST.
- The **runway entry** proposed first is the first one from the threshold you can reach **without crossing a runway**. At CYUL, runway 24L full length (R) is across 06R/24L from the terminal: the entry proposed is the first one on the terminal side; choose R in the Entry list if ATC clears you across.

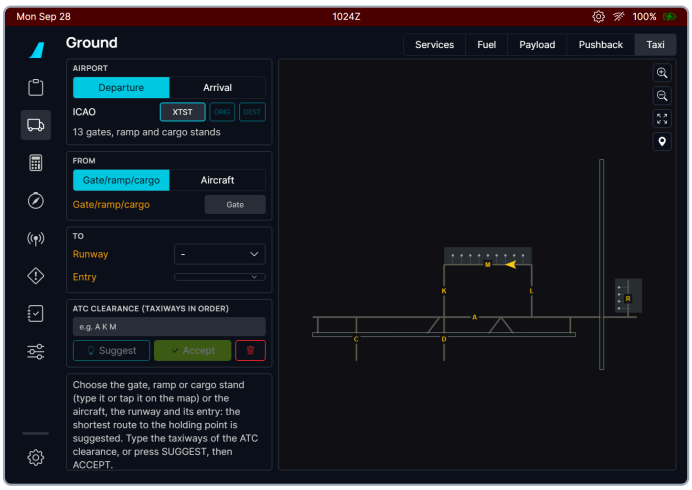


Figure 15. The whole airport, the aircraft on M.

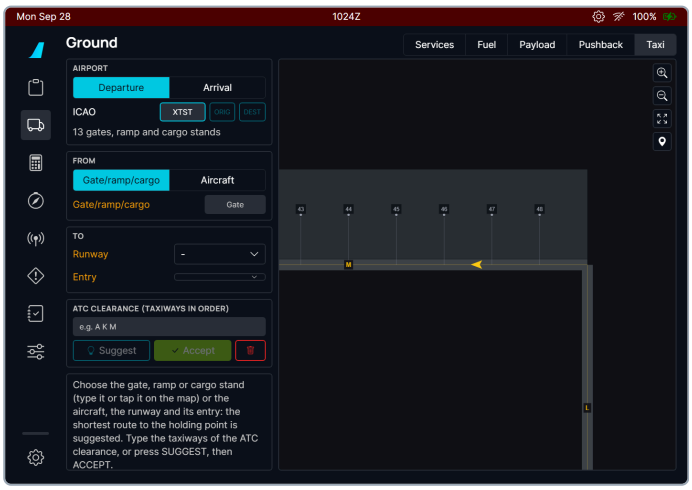


Figure 16. After the locate button: centred on the aircraft, stand names shown.

The map

- **Drag** the map to move it; **+ / -** zoom; the **frame** button fits the route (or the whole airport).
- The map is north up. Aprons are lighter grey, taxiways dark grey, runways black with a white edge. The thin yellow lines are the taxi guidance lines the route follows.
- Taxiway names are shown in yellow; the taxiways of the route in magenta. Gate names appear when you zoom in.
- **Tap a gate** on the map to choose it (when the page asks for a gate).



Figure 17. Zoomed on the apron: the gate names, the route to gate 46 along M.

What to expect

Situation	What you should see
Airport entered, nothing else	The airport map, amber labels on the inputs still missing, the help text in the route box.
Start and end chosen	The suggested route, dashed, with its clearance text, length and crossings.
SUGGEST	The clearance field holds the taxiways of the suggested route; the route is a preview (dashed).
Clearance typed	The route follows the typed taxiways in order (dashed until accepted), or the red message "No route along these taxiways, in this order".
ACCEPT	Solid line, "Accepted route" in green (A380: green flags on the OANS).
Any input changed after ACCEPT	Back to a preview: accept again.
Departure to a runway with holding position lines	The route stops at the HOLD bar before the runway.
Two entries behind the same HOLD bar (e.g. RJTT 34R C2 and C3)	The same route to the holding point: the entries only differ after it.
Gate whose lead-in is another stand's line (e.g. RJTT gate 5 behind 5R)	The route leaves along that line: "Gate 5 → Gate 5R → W ...".
Runway without holding position lines on the map	The route ends on the runway entry itself and the box says "Runway 09" instead of "Holding point RWY 09".
The route crosses a runway	Red circle with the runway name on the map, red "Crossing runway" line.
Aircraft start, aircraft far from the airport	"The aircraft is not at this airport"; no route.
Leaving the page and coming back	Everything is kept, except an Aircraft start: press Aircraft or Update again.

Limitations and why

Limitation	Why	What to do
Needs a Navigraph account.	The airport map (taxi lines, stands, holding points) exists only in the Navigraph airport database; MSFS does not give it to the aircraft.	Link your account in the flyPad settings.
Closed taxiways, one-way taxiways and NOTAMs are unknown.	The airport database has the lines painted on the ground, not their status of the day.	Type the taxiways of the ATC clearance: the route follows them.
The suggestion is the shortest route, not the usual one.	Airports have preferred flows (ATC habits, one-way loops) that are not in the data.	Always check against the clearance and the airport chart.
Some real airports have gaps in their taxi lines.	A lead-in line may stop a few metres short of the taxi line. Gaps up to 15 m between parts that are otherwise not connected are joined; larger ones are not.	If "No taxi line of the airport map connects the start to the end" appears, choose another start (the nearest taxiway, or the aircraft) or end.
Gate names follow the airport database.	The database may use names that differ from the gate signs.	Tap the gate on the map.
No route on the A380 OANS, only flags.	The A380 FCOM has no taxi route display; the flags are the FCOM way to mark points on the airport map.	-
The aircraft position is taken once.	The route is a plan to check, not a moving map that follows you.	Press Update after a change.

Troubleshooting

What you see	Why	What to do
SUGGEST and ACCEPT are greyed	There is no route yet: the gate is not found (amber label, no terminal next to the name), or the runway / entry / exit is missing.	Pick the gate from the suggestion buttons or tap it on the map; check the amber labels.
"No such name on the airport map"	No stand of the map has this name.	Tap the gate on the map.
"No taxi line of the airport map connects the start to the end"	The start and the end are on parts of the map that are not connected.	Start from the nearest taxiway (Aircraft), or choose another gate or entry.
"No route along these taxiways, in this order"	A taxiway of the clearance is misspelt, missing, or they do not connect in that order.	Check the clearance; press SUGGEST to see the names the map uses.
"Not in the Navigraph airport database"	Navigraph has no airport map for this airport.	-

How to use

1. Link your Navigraph account (flyPad > Settings > 3rd Party Options).
2. flyPad > **Ground** > **Taxi**: Departure or Arrival, the airport (or ORIG / DEST).
3. FROM: your gate (departure), the runway exit (arrival) or the aircraft; TO: the runway and its entry, or the gate.
4. Type the taxiways of the ATC clearance in order (or SUGGEST), check the route and the runway crossings, press **ACCEPT**.

How to test

#	Do this	Expected result
1	Navigraph account linked. A32NX or A380X parked at a gate of a large airport (e.g. RJTT). flyPad > Ground > Taxi, Departure, airport RJTT.	The airport map; the line under the airport gives the number of stands; amber labels on the missing inputs.
2	In FROM type your gate number.	Matching gates as buttons under the field ("16" also finds "016"); tap one, or tap the gate on the map.
3	In TO choose a runway; keep the proposed entry.	Dashed magenta route to the HOLD bar; the proposed entry is the first one reached without crossing a runway.
4	Open the Entry list.	All entries from the threshold: full length first, then the remaining length of each.
5	Press SUGGEST, then ACCEPT.	The clearance field holds the taxiways; after ACCEPT the line is solid and the box says "Accepted route".
6	Type another clearance with existing taxiways (e.g. a longer way round), without ACCEPT.	The route follows them in order, dashed, "Route along the clearance (not accepted)".
7	Type a taxiway that does not exist.	"No route along these taxiways, in this order".
8	Choose a runway the route must cross to reach.	Red circle with the runway name on the map and a red "Crossing runway" line.
9	FROM = Aircraft.	The route starts at the aircraft and goes ahead along its heading (no U-turn through the aircraft).
10	Press the locate button (pin) under the zoom buttons.	The map centres on the aircraft, stand names shown.
11	Arrival: DEST, runway, exit, gate.	Route from the exit to the gate; exits with their distance from the threshold.
12	Leave the page and come back.	Everything kept, except an Aircraft start.

2. A380X: the accepted taxi route on the OANS

> **Aircraft:** A380X | **Where:** ND (OANS) after ACCEPT on the flyPad Taxi page | **Status:** Tested in the sim by the user: OK.

What it is for

The A380 FCOM has no taxi route display: the OANS lets the crew mark points of the airport with green flags (DSC-34-10-70-20). When the crew accepts a route on the flyPad, the A380X marks it with those flags, so the route can be followed on the ND.

On the A380: the route on the OANS

When you **accept** the route on the A380 flyPad, green flags appear on the OANS (the airport map on the ND) at the start of each taxiway of the route and at the gate or holding point. They are the crew flags of the FCOM ("*to locate or mark a given point on the airport*"), so you can remove them like any flag: click one and DELETE, or ERASE ALL SYMBOLS in the ND menu. Flags you placed yourself are never touched.

- The flags appear when the OANS displays that airport (they come back if the OANS reloads it).
- A new accepted route replaces the flags of the previous one; a route that is no longer accepted removes them.
- The A320 has no OANS: nothing else changes on it.

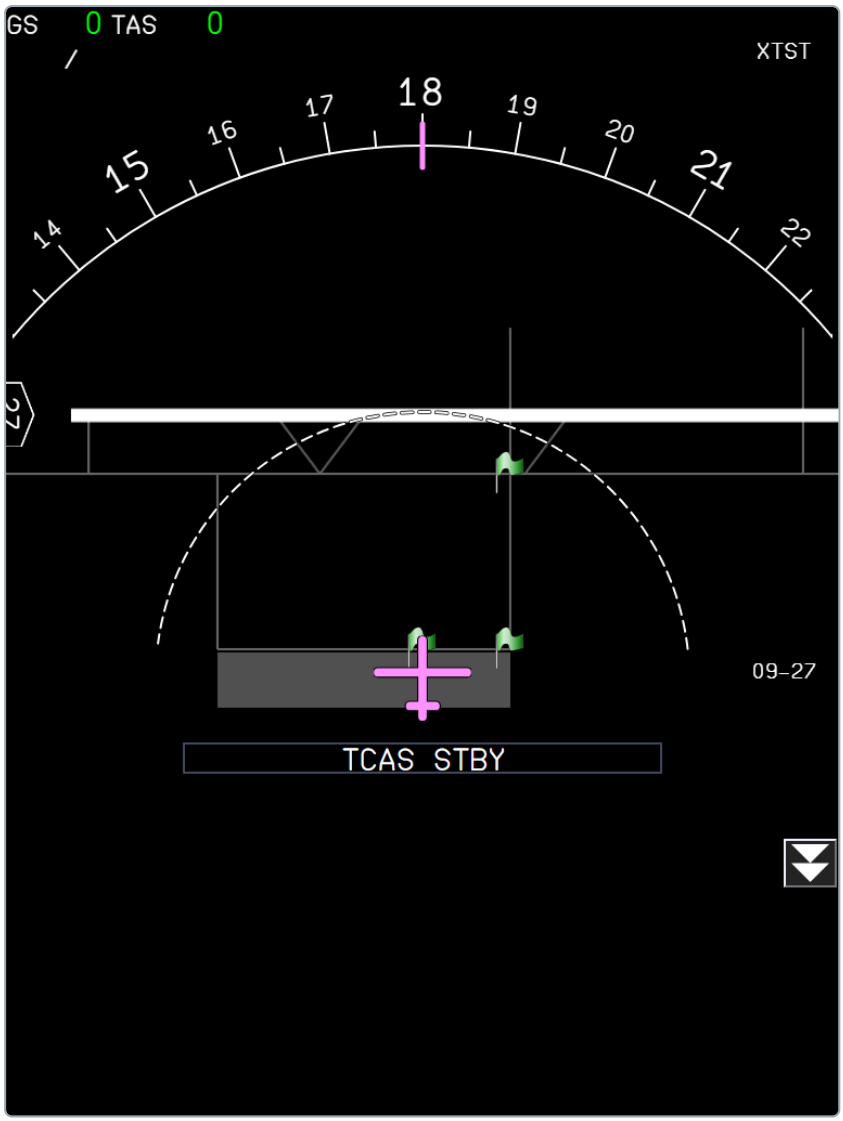


Figure 18. The A380 ND (OANS, ARC, zoom range) at gate 42 facing south, after ACCEPT of the departure route above (gate 42 to runway 09): green flags where M, K and A start (S1 and the holding point are further west, off the picture). Harness picture: the test airport has no buildings.

> **NOTE**
> Tested in the sim by the user: OK: the flags are set on ACCEPT and removed when the route changes.

Limitations and why

Limitation	Why	What to do
Flags only, no route line.	The A380 FCOM has no route display on the OANS; flags are its way to mark points.	-
An Aircraft start is lost when leaving the Taxi page, so its flags go away when you come back.	The aircraft start is local to the page.	Select Aircraft again and ACCEPT.

How to use

1. Accept a route on the flyPad Taxi page (A380X).
2. On the ND, select the OANS (ARC or ROSE, a zoom range) at that airport: the flags show the start of each taxiway and the end of the route.

How to test

#	Do this	Expected result
1	A380X at a gate, Navigraph linked, OANS showing the airport on the ND. flyPad Taxi page: route to a runway, **ACCEPT**.	Green flags on the OANS at the start of each taxiway of the route and at the holding point; the flyPad status says "green flags on the OANS".
2	Place your own flag on the OANS, then change the clearance on the flyPad (route no longer accepted).	The route flags go away; your own flag stays.
3	ACCEPT a new route.	Its flags replace those of the previous route.
4	Click a route flag on the OANS and DELETE, or ERASE ALL SYMBOLS in the ND menu.	The flags are removed like crew flags.

3. Airport ATC frequencies on the taxi page

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Taxi, bottom left of the map | **Status:** Tested in the sim by the user: OK.

What it is for

Before taxi the crew needs the ground, tower and delivery frequencies of the airport. The taxi page shows them in a box laid out like the frequency box of an airport chart, for the origin, the destination or the airport of the page. The Navigraph airport map has no frequencies: they come from the sim's own airport data.

What you see

- A green **Frequencies** button at the bottom left of the map; it opens the **FREQUENCIES** box (X closes it).
- Airport buttons at the top: **ORIG** and **DEST** of the SimBrief flight plan loaded in the flyPad, and **Airport** when the page airport is another one.
- One row per chart label, in the chart order: ATIS, AWOS, ASOS, CTL, APP, DEP, TWR, GND, DLV, CTAF, UNICOM, MULTICOM, FSS; the frequencies in MHz, two columns, without duplicates, in the order of the sim data. Other types (ILS...) are left out.
- Messages: "Enter an airport", "Loading the frequencies...", "Not in the sim's airport data", "No frequency for this airport in the sim's airport data".

Limitations and why

Limitation	Why	What to do
The list is the sim's, not the chart's.	The chart's selection of frequencies is in no data the aircraft can read; MSFS lists all its frequencies.	Check the chart for the one to use.
No station names, no tuning.	The box is a reference, like the chart box.	Tune the RMP as usual.

How to use

1. On the Taxi page, press **Frequencies** (bottom left of the map).
2. Choose ORIG, DEST or the page airport; read the frequency you need.

How to test

#	Do this	Expected result
1	SimBrief OFP loaded in the flyPad (e.g. RJTT to another airport). Ground > Taxi, airport RJTT. Press **Frequencies**.	The FREQUENCIES box with ORIG RJTT selected and DEST; rows ATIS, APP, DEP, TWR, GND, DLV... with MHz values (three decimals).

2	Press DEST.	The frequencies of the destination.
3	Change the page airport to a third airport.	An "Airport" button appears and is selected; its frequencies.
4	Compare with the airport chart or the sim world map.	The same frequencies (the sim may list more).
5	Press X.	Back to the Frequencies button.

4. Services page linked to GSX

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Services, GSX panel | **Status:** First version tested in the sim by the user with GSX 4 (A380X, RJTT): four bugs found. Their fixes on this branch (panel position, counters, greyed switch, status lines) are not sim-tested on this classic page. The Material 3 Services page of `core/efb-redesign`, which carries them with further GSX fixes, was tested in the sim (A32NX: 170 passengers given to GSX, count 0 to 170, Done).

What it is for

With GSX installed, the ground services are GSX's, but the flyPad Services buttons called the sim's own vehicles. A **GSX services** switch now links the page to GSX through the GSX Remote API (version 2, local server on port 8744): the service buttons request the GSX services and show what GSX is doing. With the switch off the page works as before.

What you see

- The **GSX panel** under the right-hand group of buttons (GPU, cargo door): the switch and a status line: "GSX not found: start GSX, with its Remote control server on (GSX settings)" (switch greyed), "GSX is starting...", "GSX found" (switch off) or "Service buttons linked to GSX, GSX menus and messages shown here".
- Linked: **Jet Bridge**, **Stairs**, **Fuel Truck**, **GPU** and **Catering Truck** request the GSX services. The **Baggage Truck** button is hidden: GSX loads the bags during boarding.
- The panel adds the services without an aircraft button: Deboarding, Boarding, Pushback (Departure), De-ice, Water, Lavatory, Cleaning, named as GSX names them.
- Under each linked button a status line: *On its way*, *Waiting*, GSX's own phase (or *In progress*), *Leaving*, *Done*, with its count; a progress bar for the fuel truck from the "x / y kg" refuelling status. Button colours follow the service: amber when called or waiting, green while performed, amber and locked while leaving, dim when GSX cannot serve it.
- Linking switches on the flyPad **GSX Fueling** and **GSX Payload Synchronization** (Settings > 3rd Party Options), so GSX boards the passengers and loads the fuel set on the Payload and Fuel pages.
- While linked, GSX is under remote control: no GSX pop-up menu or notification in the sim. GSX's questions (menus that stay open, e.g. the choice of operator) and its messages appear on the panel; tap an answer, or "Close the GSX menu".
- A request GSX refuses (e.g. boarding before the doors are open) gives the message "GSX did not accept <service>" with GSX's reason.

Limitations and why

Limitation	Why	What to do
Needs GSX with its Remote control server on.	The link uses the GSX Remote API (GSX 4).	Turn on the Remote control server in the GSX settings.

The four bugs of the first sim test (panel over the fuselage drawing, Board counter stuck at 0, passenger count not moving, switch not greyed without GSX) are fixed here but not checked in the sim on this page.	The fixes came after the test. `core/efb-redesign` reworks the counter (the flyPad counts the GSX passengers itself and gives GSX its passenger number) and the greyed switch again; that version is the one tested in the sim.	Merge `core/efb-redesign`; report any of them if seen again.
Turning the link off leaves the GSX fuel and payload syncs on.	Only linking changes them.	Turn them off in Settings > 3rd Party Options if wanted.
The page keeps the classic FBW look on this branch.	The Material 3 redesign of the Services page is in `core/efb-redesign`.	-

How to use

1. Start GSX with its Remote control server on.
2. flyPad > **Ground** > **Services**: when the panel says "GSX found", turn on **GSX services**.
3. Request the services with the buttons and the GSX panel; answer GSX's questions on the panel.
4. Turn the switch off to go back to the FBW services.

How to test

#	Do this	Expected result
1	A380X (or A32NX) at a gate, GSX not running. flyPad > Ground > Services.	GSX panel under the GPU / cargo door group; switch greyed; amber "GSX not found: start GSX, with its Remote control server on (GSX settings)".
2	Start GSX with the Remote control server on.	"GSX is starting...", then "GSX found"; the switch becomes active.
3	Turn on **GSX services**.	Green "Service buttons linked to GSX..."; the Baggage Truck button disappears; the panel lists Deboarding, Boarding, Pushback, De-ice, Water, Lavatory, Cleaning; Settings > 3rd Party Options shows the GSX fuel and payload syncs on.
4	Press **Jet Bridge** (or Stairs).	Amber, status "On its way", then green while performed, "Done" at the end. No GSX menu pops up in the sim.
5	Press **Fuel Truck**.	Status and progress bar following "x / y kg"; the fuel set on the Fuel page.
6	Press **Boarding** on the panel (doors open).	The count of boarded passengers rises to the number set on the Payload page.
7	When GSX asks a question (e.g. the operator choice), answer it on the panel.	GSX continues with that answer.
8	Turn the switch off.	The Baggage Truck button is back; the buttons call the FBW services again.

5. GSX external power sync when it is enabled

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Settings > 3rd Party Options > GSX External Power Synchronization | **Status:** Tested in the sim by the user (A380X, GSX 4, RJTT): EXT AVAIL on the ELEC panel with the GSX GPU, OK.

What it is for

The FBW option *GSX External Power Synchronization* makes the external power available (`L:A32NX_EXT_PWR_AVAIL`) when the GSX GPU is connected or the jetway is requested or connected, until pushback. It only looked at GSX when a GSX state changed: enabling the option with the GPU or the jetway already connected did nothing until the next GSX change. It now checks GSX at once when it is enabled.

What you should see

Situation	What you should see
Option enabled with the GSX GPU or jetway already connected	External power available at once: EXT AVAIL on the overhead ELEC panel.
Option enabled before GSX connects	As before: available when the GPU connects.

Limitations and why

Limitation	Why	What to do
With GSX 4, EXT AVAIL may already show before the option is enabled.	GSX 4 also sets the sim's own external power variable.	Watch `L:A32NX_EXT_PWR_AVAIL:1` with a SimVar tool to see the sync itself.

How to use

1. flyPad > **Settings** > **3rd Party Options**: turn on **GSX External Power Synchronization**, before or after GSX connects its GPU or jetway.
2. Press the EXT PWR pushbutton when AVAIL shows.

How to test

#	Do this	Expected result
1	A380X (or A32NX) cold and dark at a GSX gate, the option OFF. Call the GSX GPU (or the jetway) and wait until it is connected.	`L:A32NX_EXT_PWR_AVAIL:1` stays 0 (unless GSX 4 already made EXT AVAIL show, see Limitations).
2	Turn on **GSX External Power Synchronization**.	At once: `L:A32NX_EXT_PWR_AVAIL:1` = 1, EXT AVAIL on the overhead ELEC panel (before: nothing until the next GSX state change).
3	Press EXT PWR.	The aircraft is powered by the ground power.
4	Request the GSX pushback.	The external power becomes unavailable, as before.

6. Taxi route along real ATC clearances

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Taxi, clearance field | **Status:** Not sim-tested (added after the user's RJTT report of the same day; unit tests in taxiRoute.spec.ts "Taxi route along an ATC clearance", on a synthetic RJTT-like map).

What it is for

At RJTT the clearance `W A E C1 C1` typed from the aircraft after pushback gave "No route along these taxiways", and a full BeyondATC instruction ("via L4 cross runway 34L, taxi to holding point C1, runway 34R, via L4, A4B, W4, W, A, E, C1") could not be used. Real clearances leave out the apron taxiways, repeat the holding point, name runway exits and come as sentences. The route finder now accepts them:

- **Typed list or pasted instruction:** the clearance field reads a list of taxiways or a pasted ATC instruction. Phraseology words (VIA, TAXI, TO, HOLDING POINT...), runway designators, the stand after STAND or GATE and frequencies are left out. The runways after CROSS (RUNWAY) are the cleared crossings. The clearance limit (after TO or HOLDING POINT) is the last taxiway. HOLD SHORT (OF) a taxiway is left out. "A-4B" is read as the airport name A4B. A name said twice in a row is one taxiway.
- **Apron taxiways before the clearance:** before the first cleared taxiway, taxiways that are not in the clearance may lead to it, within 2000 m of the start. They cost more (x1.5), so the route joins the clearance as soon as it can. A cleared taxiway is never used out of order.
- **Uncleared runway crossings:** in that first part, a runway crossing that the clearance does not mention costs a large penalty: the route goes round when it can; the crossing still shows on the map. "cross runway 34L" in the clearance removes the penalty.
- **Runway exit lines** named in the clearance are cleared taxiways; a clearance that starts with the landing exit ("B1 A K M") works. Exit lines are listed in the route taxiways (SUGGEST), except the start exit.
- **Unknown names:** the page lists the cleared names that are not a taxiway of the airport map, under the clearance field in amber ("Not a taxiway of the airport map: X9"). The field placeholder now says "e.g. A K M, or paste the ATC instruction".

Situation	What you should see
From the aircraft after pushback, clearance `W A E C1 C1`	A route: the apron taxiways from the aircraft to W are added, then W A E C1.
A full ATC instruction pasted in the field	The taxiways in order (for the RJTT example L4 A4B W4 W A E C1); "Crossing runway 34L" shown if the route crosses it.
A name in the clearance that the airport map does not have	"Not a taxiway of the airport map: <name>" under the field.
A clearance taxiway used out of order would be shorter	Not used: the clearance order is kept.

Limitations and why

Limitation	Why	What to do
The RJTT taxiway names A4B, W4 and L4 are not checked against the real Navigraph map.	No real airport map data offline: the tests use a synthetic map.	Report any name listed as "Not a taxiway of the airport map".
The apron taxiways before the clearance are searched within 2000 m of the start only.	Keeps the route from wandering on other taxiways.	Add the first taxiway to the clearance if it is further.

How to use

1. flyPad > Ground > Taxi: set FROM (gate, exit or Aircraft) and TO as before.
2. Type the cleared taxiways in order, or paste the whole ATC instruction (for example from BeyondATC) in the clearance field.
3. Check the route, any "Crossing runway" line and any "Not a taxiway of the airport map" line, then ACCEPT.

How to test

#	Do this	Expected result
1	RJTT, Departure, FROM Aircraft after pushback near gate 67, TO runway 34R entry C1. Type `W A E C1 C1`.	A route (not "No route along these taxiways"), with the first taxiways added from the aircraft.

2	Paste the full instruction "via L4 cross runway 34L, taxi to holding point C1, runway 34R, via L4, A4B, W4, W, A, E, C1".	Route L4 A4B W4 W A E C1; "Crossing runway 34L" shown if L4 crosses it.
3	Type a clearance with a name that does not exist (for example X9).	"Not a taxiway of the airport map: X9" in amber under the field.
4	Arrival: clearance starting with the landing exit (for example `B1 A K M` at an airport that has them).	A route from the exit along the clearance.

What PR testers should check

- At RJTT (Departure, From Aircraft after pushback near gate 67, runway 34R entry C1): "W A E C1 C1" gives a route with the first taxiways added from the aircraft.
- The pasted BeyondATC instruction gives L4 A4B W4 W A E C1, with "Crossing runway 34L" if L4 crosses it.
- Note any name listed as "Not a taxiway of the airport map".
- Not sim-tested yet.

7. GSX power sync: the setting is re-read, so enabling it takes effect

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Settings > 3rd Party Options > GSX External Power Synchronization | **Status:** Not sim-tested (added; unit tests: GsxSync.spec.ts, 3, all failing without the change).

What it is for

Follow-up of the feature above. The flyPad switch stores the setting and signals the change through a listener that the extras host (the code that runs the GSX sync) never registers, so the change did not always reach it: turning the sync on with the GSX GPU or jetway already connected could still leave EXT PWR unavailable. The extras host now re-reads the stored setting (at most once a second) and checks the GSX power source again when the setting goes from off to on.

Limitations and why

Limitation	Why	What to do
Up to one second between the switch and EXT AVAIL.	The setting is polled once a second.	-

How to use

1. As above: turn on GSX External Power Synchronization, before or after GSX connects its GPU or jetway.

How to test

#	Do this	Expected result
1	Cold and dark at a GSX gate, sync OFF, GSX GPU connected.	No EXT AVAIL from the sync.
2	Turn the flyPad GSX power sync on.	EXT PWR AVAIL within a second.

What PR testers should check

- Turn the flyPad GSX power sync on with the GSX GPU connected: EXT PWR AVAIL at once.
- Not sim-tested yet.

8. External power follows the sim's ground power (GSX GPU, sim GPU, jetway)

> **Aircraft:** A32NX and A380X | **Where:** Overhead ELEC panel EXT PWR; flyPad > Ground > Services GPU button | **Status:** Not sim-tested (added; unit tests: GPUManagement.spec.ts, 10, failing first).

What it is for

The aircraft read the sim's external power only on the first frame and when the flyPad GPU button was used. A GPU or a jetway connected after the flight loaded (GSX 4, the sim GPU) did nothing until the GPU door moved. Now the external power follows the sim's ground power, whoever connects it.

Situation	What you should see
Sim ground power appears (GSX GPU, sim GPU or jetway power), aircraft at or below 0.3 kt	EXT PWR AVAIL on the overhead (if the external power was not already available).
That sim ground power goes	EXT PWR no longer available, but only if it was connected from the sim power.
Crew connection from the flyPad GPU button, or EXT PWR set by the GSX sync	Kept: the sim power going does not remove it.

Limitations and why

Limitation	Why	What to do
Not above 0.3 kt.	A moving aircraft cannot be on ground power.	-

How to use

1. Connect the ground power any way (GSX GPU or jetway, the sim GPU, or the flyPad GPU button) and press EXT PWR when AVAIL shows.

How to test

#	Do this	Expected result
1	Load the flight cold and dark at a gate, then call the GSX GPU (or connect the jetway).	EXT AVAIL on the overhead as soon as the GPU is connected (no wait for the GPU door).
2	Disconnect the GSX GPU.	EXT AVAIL goes.
3	Connect the GPU with the flyPad Services GPU button, then make the sim power go.	EXT AVAIL stays (crew connection kept).

What PR testers should check

- GSX GPU and jetway connect / disconnect after the flight is loaded.
- The flyPad GPU button.
- Not sim-tested yet.